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22ESEC21

SECOND SEMESTER B.Tech EXAMINATION, APRIL - 2023

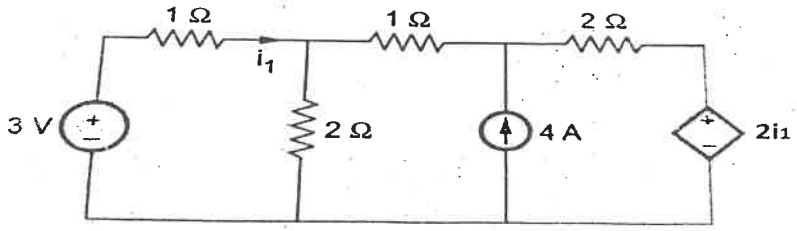
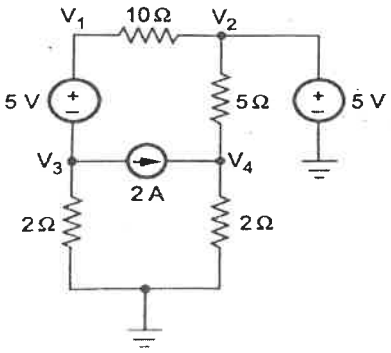
ELECTRIC CIRCUITS

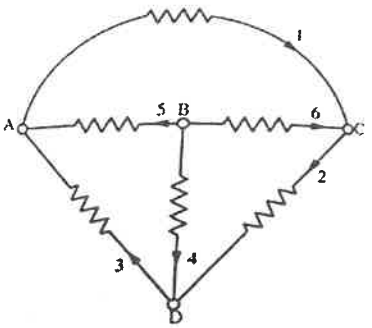
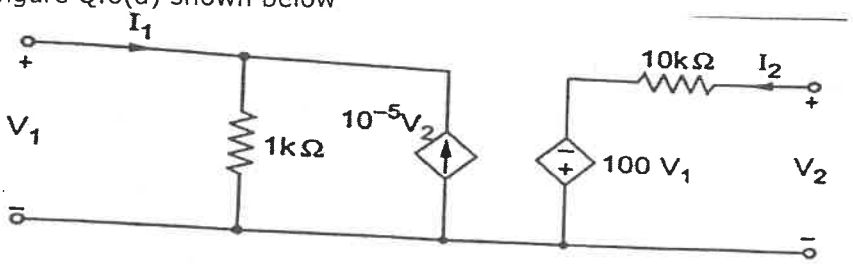
Time: 3 Hours **Maximum Marks: 100**

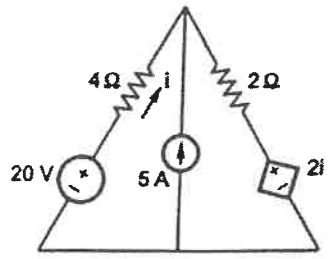
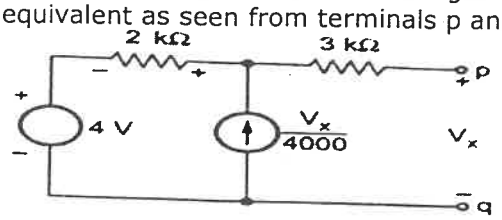
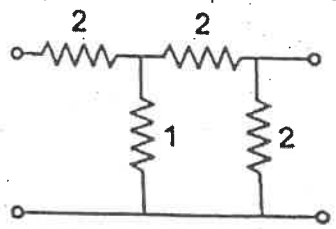
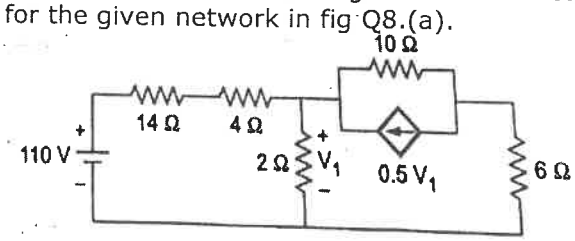
Instructions: i. Missing data may be suitably assumed.

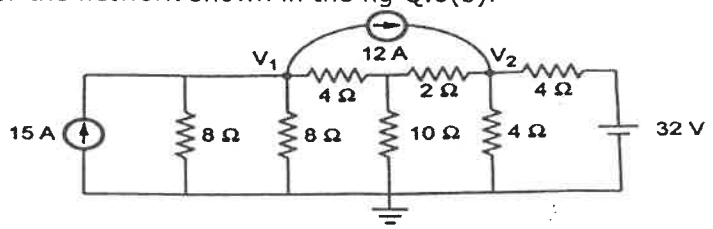
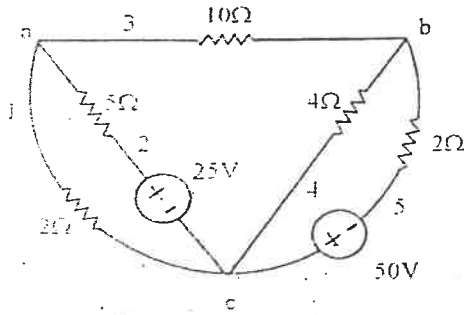
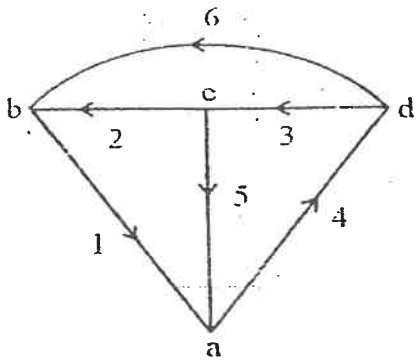
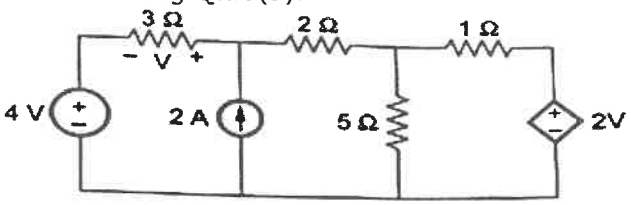
ii. Answer All Questions

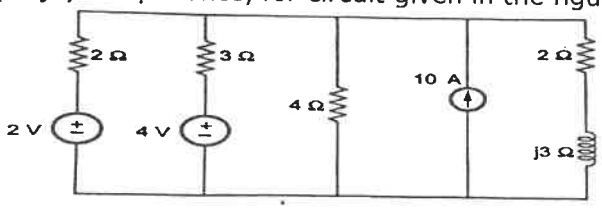
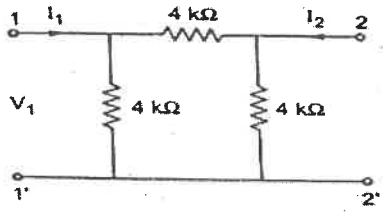
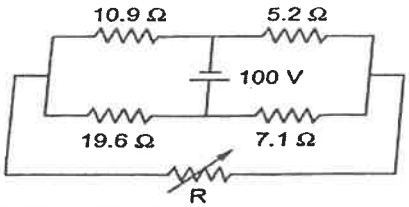
PART-A **05 X 02=10**

	a.	Write the procedure for Loop analysis.	L-1	CO1	02
	b.	Define a graph and a tree in network topology.	L-1	CO2	02
	c.	Define superposition principle. Why it cannot be applied to power.	L-1	CO3	02
	d.	Define i) Figure of merit. ii) selectivity of resonant circuit	L-1	CO5	02
	e.	Define z-Parameters and h-Parameters.	L-1	CO6	02
		PART-B	**06 X 05=30**		
2.	a.	Determine the current in 2Ω resistor by mesh analysis in the figure Q.2(a)  Fig Q.2(a)	L-3	CO1	05
		OR			
	b.	For the network shown in the figure Q.2(b) determine the node voltage V_1, V_2, V_3 and V_4 using nodal analysis.  Fig Q.2(b)	L-3	CO1	05

3.	a.	Draw the graph, choose a tree and then obtain the tie-set matrix for the figure Q.3(a) shown below  Fig Q.3(a)	L-2	CO2	**05**
		OR			
	b.	For the incidence matrix shown below, draw the related graph. $A_a = \begin{bmatrix} +1 & 0 & 0 & 0 & +1 & -1 \\ -1 & +1 & +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 & 0 & +1 \\ 0 & 0 & -1 & +1 & -1 & 0 \end{bmatrix}$	L-2	CO2	**05**
4.	a.	State and prove superposition theorem.	L-3	CO3	**05**
		OR			
	b.	State and prove Millman's theorem.	L-3	CO3	**05**
5.	a.	It is required that a series RLC circuit should resonate at 1MHz. Determine values of R, L and C if bandwidth of the circuit is 5 kHz and its impedance is 50Ω at resonance.	L-2	CO5	**05**
		OR			
	b.	In a circuit a capacitor is connected in parallel with a series combination of resistor and inductor. If R = 25Ω, L = 0.5H and C=5μF , Determine the ω_{ar}, Q and bandwidth.	L-2	CO5	**05**
6.	a.	Determine the [z] and [y] for the two port network shown in the figure Q.6(a) shown below  Fig Q.6(a)	L-3	CO6	**05**
		OR			

	b.	For the circuit shown in figure Q.6(b) solve for the current i using superposition theorem  Fig Q.6(b)	L-3	CO6	**05**
7.	**a.**	For the network shown in the figure Q.7(a) obtain the Thevenin's equivalent as seen from terminals p and q.  Fig Q.7(a)	L-3	CO3	**05**
		OR			
	b.	For the network of figQ.7(b) Determine the Z-parameters. Also draw the z-parameter equivalent circuit.  Fig Q.7(b)	L-3	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Solve for the current through 4Ω and 6Ω resistors using mesh analysis for the given network in fig Q8.(a).  Fig Q.8(a)	L-3	CO1	**10**
		OR			

	b.	Solve for the power dissipated in 10Ω resistor by node voltage method for the network shown in the fig Q.8(b).  Fig Q.8(b)	L-3	CO1	**10**
9.	**a.**	For the network in figure Q.9(a), draw the graph. Select 2 and 4 as tree branches. Draw the tie-set matrix. Write down the equilibrium equations with loop currents as variables. Solve these equations and find the various branch voltages and currents. The integers indicate branch numbers.  Fig Q.9(a)	L-3	CO2	**10**
		OR			
	b.	Write the number of possible Trees of a graph shown in the figure Q.9(b).  Fig.Q.9(b)	L-3	CO2	**10**
10.	**a.**	Solve for 'V' using the principle of Superposition for the given network shown in the fig Q.10(a).  Fig Q.10(a)	L-3	CO4	**10**
		OR			

	b.	Use Millmans theorem to solve for the current flowing through (2+j3)Ω impedance, for circuit given in the figure Q.10(b).  Fig Q.10(b)	L-3	CO4	**10**
11.	**a.**	Two coils; one of R_1 = 0.51Ω, L_1 = 32mH and other coil of R_2 = 13Ω, L_2=15mH are in series and are in series with a capacitor of 25μF and 62μF and a series resistor of resistance 0.24Ω . Determine the Resonant frequency, Q-factor , Bandwidth, Power dissipated in the circuit at resonant frequency.	L-3	CO5	**10**
		OR			
	b.	A series RLC circuit has R = 4Ω , L = 1mH and C = 10μF , calculate Q-factor, bandwidth, resonant frequency and the half power frequencies f_1 and f_2.	L-3	CO5	**10**
12.	**a.**	Determine the y-parameters for the circuit shown in the fig Q.12(a). Then use the parameter relationship to find the ABCD parameter.  Fig Q.12(a)	L-3	CO6	**10**
		OR			
	b.	Explain h and y-parameters. Also derive expressions for [h] in terms of [y].	L-2, L-3	CO6	**10**
13.	**a.**	Determine the value of R that will receive maximum power for the circuit shown in the figure Q.13(a) shown below. Determine the power.  FigQ.13.(a)	L-3	CO4	**10**
		OR			
	b.	Show that resonant frequency of a series resonance circuit is equal to the geometric mean of two half power frequencies.	L-3	CO5	**10**